

Block 6

Operational Aspects of Financial Markets

Unit 18	Efficient Portfolio Frontier
Unit 19	Capital Asset Pricing Model
Unit 20	Arbitrage Pricing Theory
Unit 21	Pricing of Derivatives
Unit 22	Corporate Finance
Unit 23	Foreign Direct Investment and Foreign Portfolio Investment

UNIT 18 EFFICIENT PORTFOLIO FRONTIER

Structure

18.0 Objectives

18.1 Introduction

18.2 Portfolio Management

18.2.1 Stages of Portfolio Management

18.2.2 Benefits of Portfolio Management

18.3 Relationship between Risk and Return

18.3.1 Risks Involved in Investments

18.3.2 Measuring Risks

18.3.3 Expected Returns

18.4 Valuation of Portfolio and Expected Returns from a Portfolio

18.4.1 Valuation of Portfolio

18.4.2 Expected Return and Risk from a Portfolio

18.5 Markowitz Portfolio Theory

18.5.1 Assumptions of the Markowitz Model

18.5.2 Opportunities Set/ Feasible Set of Portfolios

18.5.3 Efficient Set of Portfolios

18.5.4 Criticisms of the Efficient Frontier

18.6 Let Us Sum Up

18.7 Key Words

18.8 Some Useful Books and References

18.9 Answers/Hints to Check Your Progress Exercises

18.0 OBJECTIVES

After going through this unit, you will be able to:

- describe the importance of diversification;
- explain the need for portfolio management;
- calculate the expected return and valuation of the portfolio;
- explain the relationship between risk and return;
- critically examine Markowitz's Portfolio theory; and
- explain Efficient Portfolio frontier.

18.1 INTRODUCTION

In the realm of finance, the adage “Don't put all your eggs in one basket” holds significant importance. This principle underscores the fundamental strategy of spreading investments across various assets and asset classes to mitigate risk and optimize returns.

Diversification is the practice of spreading investments across various assets and asset classes to mitigate risk and optimize returns. It means creating a well-balanced portfolio, wherein a range of investments is judiciously selected and blended to counteract the impact of volatility in any single investment.

The need for diversification stems from the volatility inherent in financial markets. Given that certain investments may thrive while others may decline due to various factors such as economic recessions or geopolitical unrest, diversification serves as a strategy for investors to mitigate the impact of adverse market fluctuations. By spreading their investments across different assets, investors seek to protect their wealth from significant losses in the long run.

Diversification is a key principle within portfolio management. A portfolio represents a collection of investments held by an individual or entity, comprising a broad range of assets including stocks, bonds, mutual funds, real estate, and other financial instruments.

Many people find it difficult to make effective investment decisions because of uncertainty in the market, conflicting investment goals and sometimes even due to heavy information load. In response to these challenges, numerous theories in portfolio management have endeavoured to provide solutions. They have tried to answer questions like how to diversify. Which financial assets to invest in? How to combine the assets into a portfolio? What is the portfolio's risk? how to calculate the expected return? etc.

In this unit, we will discuss portfolio management, expected return and valuation of portfolio, relationship between risk and return. We will also discuss Markowitz's Portfolio theory and Efficient Portfolio frontier. In subsequent units, we will talk about the Capital asset pricing model (CAPM) and Arbitrage pricing theory (APT).

18.2 PORTFOLIO MANAGEMENT

A portfolio is essentially a collection of investments owned by an individual or entity, which can include various types of assets like stocks, bonds, mutual funds, and real estate. However, it's more than just a bunch of investments grouped; it's a strategic approach to managing wealth, minimizing risks, and achieving financial goals.

Portfolio management is the practice of overseeing and optimizing these investment collections. It involves carefully deciding how to allocate resources, balancing risk and returns, and adapting to changes in the market. This includes activities such as deciding which assets to invest in, evaluating the level of risk involved, and monitoring how well the portfolio is performing.

The main goal of portfolio management is to create portfolios that match the specific financial goals, risk tolerance, and timeframes of investors. Whether someone wants to preserve capital, generate income, or increase wealth, portfolio managers develop strategies tailored to these objectives. They do this through thorough analysis and smart decision-making, aiming to achieve the best possible returns while minimizing the chances of losses.

Additionally, portfolio management is an ongoing process that requires constant attention and adjustment. Changes in the market shifts in the economy, and alterations in investors' situations mean that portfolios need to be reviewed and adjusted regularly. This means that successful portfolio management relies on a combination of financial expertise, careful research, and proactive decision-making.

18.2.1 Stages of Portfolio Management

Portfolio management is a dynamic procedure that includes the following fundamental stages:

- Understanding the goals, constraints, and preferences of investors to create an investment policy.
- Creating and executing strategies aligned with the formulated investment policy.
- Choosing asset classes and specific securities within each class based on their risk-return characteristics.
- Regularly assessing and tracking the portfolio's performance by consistently evaluating market conditions and company performance.
- Analyzing the portfolio's outcomes to compare against predetermined targets and making necessary adjustments for future enhancements.

18.2.2 Benefits of Portfolio Management

- 1) **Consistent Risk Measurement:** It ensures that risks are measured in the same way every time, providing a clear understanding of potential risks.
- 2) **Considering Correlation and Diversification:** It takes into account how different parts of the organization or investment options relate to each other. This understanding can help in reducing overall risk.

- 3) **Guidance for Strategic Planning:** It provides direction for the company's plans, highlighting areas where improvements are needed in terms of risk and return.
- 4) **Increased Transparency in Decision Making:** Portfolio management helps make decisions clearer by consistently evaluating all business units and options.
- 5) **Understanding Value Creation:** Portfolio management helps in better grasping how new investment opportunities can create value for the organization.
- 6) **Identifying Improvement Areas:** Portfolio management helps identify where the company needs to focus to enhance its risk-return position.
- 7) **Considering Non-Financial Implications:** It acknowledges the qualitative and non-financial impacts of decisions, not just the financial aspects.

Check Your Progress 1

- 1) What is portfolio management? What are the different stages of portfolio management?
.....
.....
.....
- 2) Why is portfolio management important for an investor?
.....
.....
.....

18.3 RELATIONSHIP BETWEEN RISK AND RETURN

Most investors are risk-averse. This does not mean that people will not take a risk. That means they will take a risk only if they expect to be adequately rewarded for taking it. People have different degrees of risk aversion. Some are more willing to take a chance than others. Choosing an investment management strategy involves weighing the trade-off between risk and return. Active management might yield higher returns but comes with a greater risk of not meeting the portfolio's required return. Conversely, passive management offers market-based returns in diversified portfolios, reducing risky bets but potentially limiting excess returns. When deciding between passive and active management, it's crucial to determine if the goal is to solve a cash flow problem over time or to outperform the market.

18.3.1 Risks Involved in Investments

Total risk refers to the overall variability of the returns of a financial asset. It combines both diversifiable and non-diversifiable risks and represents the overall uncertainty and potential for loss associated with an investment. Recognizing the types of risks allows investors to implement appropriate risk management strategies. Understanding the different types of risks helps investors construct well-balanced portfolios. By diversifying across assets and sectors, investors can reduce the impact of specific risks on their overall portfolio performance.

The following types of risks are faced in the market:

Diversifiable risk, also known as unsystematic risk (unique risks), refers to firm-specific risks that affect individual stock prices rather than the entire industry or sector.

The key components of diversifiable risk are business, financial, and management risks. Diversifiable risk is crucial for investors seeking better returns and protecting their initial investment.

An uncertain risk can arise from various factors such as scams, labour strikes, regulatory penalties, management changes, internal issues, or news specific to the firm. Diversifiable risk can be mitigated through diversification, which involves spreading investments across different assets or sectors. By diversifying, investors can reduce the impact of adverse events on any single investment.

Systematic risks, or non-diversifiable risks, are the risks that affect the overall market or a specific sector. It is influenced by factors that impact the entire economy, such as macroeconomic indicators, market sentiment, or geopolitical events. Systematic risks are inherent risks that exist in the stock market. They're also called "non-diversifiable risks" or "market risks" since they impact the entire asset class.

Non-diversifiable means that an organization can't control, minimize, or avoid systematic risks. These risks are typically due to various external factors like the current geopolitical situation, monetary policy, and natural disasters. Non-diversifiable risk cannot be eliminated through diversification because it is inherent in the market itself. Investors are exposed to non-diversifiable risk regardless of how diversified their portfolios are. The only way to manage non-diversifiable risk is through strategies such as asset allocation or hedging. For example, COVID-19 is a systematic risk because it impacted the entire stock market.

Diversifiable risk analysis is essential for investors and portfolio managers as it helps in understanding and managing the risks associated with individual securities or assets. By assessing and diversifying their investments, investors can reduce the impact of firm-specific risks and potentially improve their risk-return profile. Diversifiable risk analysis enables investors to make

informed decisions about portfolio allocation, asset selection, and risk management strategies to achieve their investment objectives while mitigating unnecessary risks.

18.3.2 Measuring Risks

The risk linked to a stock pertains to the fluctuation in its rate of return. Investors need to quantify and assess this risk accurately. The risk probability distribution, depicting potential returns on investments, represents the total risk associated with an investment.

Variance

Variance takes into account the derivation of all possible returns from their mean or expected value. the formula used for variance is as follows:

$$\sigma^2 = \sum_{i=1}^n [R_i - E(R)]^2 X P$$

Where R_i refers to possible returns when $i=1$ to n , $E(R)$ is the expected value of the investment and p is the probability of the return.

Standard deviation can be calculated as the square root of variance, i.e.:

$$\sigma = \sqrt{\sum_{i=1}^n [R_i - E(R)]^2 X P}$$

18.3.3 Expected Returns

Expected return denotes the anticipated profit or loss an investor can reasonably foresee from an investment within a specific timeframe. It is an important factor that determines portfolio selection. To ascertain return, various factors like market trends, economic predictions and historical data are analysed.

The concept of expected return is useful because there is uncertainty about the return on investment. Thus, an expected return means the average return that an investor expects to receive on investment over some time. It is calculated by the sum of each possible return multiplied by its probability. The expected return is calculated by using the following formula:

$$E(R) = \sum_{i=1}^n R_i P_i$$

Where:

$E(R)$ =expected return on an investment

n =number of possible terms

R_i = the i^{th} possible return

P_i = the probability of i^{th} return R_i

Example 18.1: Mr X has invested an Equal amount in Security A and Security B. The expected return during boom and depression with equal probability of occurrence is as follows:

Economic condition	Expected return	
	Security A	Security B
Boom	15	12
Depression	5	7

Calculate the Expected return and standard deviation of each security.

Solution: Calculation of expected rate of return

Economic condition	Expected return of		Probability	Expected returns	
	Security A	Security B		Security A	Security B
Boom	15	12	0.5	7.5	6
Depression	5	7	0.5	2.5	3.5
Expected return (ER)				11.00	9.5

Calculation of standard deviation of Security A

Economic condition	R_A	$R_A - E(R)$	$(R_A - E(R))^2$	Probability	$(R_A - E(R))^2 \times P$
Boom	15	4	16	0.5	8
Depression	5	-6	36	0.5	18
Variance					26

Standard deviation of security A = $\sqrt{26} = 5.09$

Calculation of standard deviation of Security B

Economic condition	R_B	$R_B - E(R)$	$(R_B - E(R))^2$	Probability	$(R_B - E(R))^2 \times P$
Boom	12	2.5	6.25	0.5	3.125
Depression	7	-2.5	6.25	0.5	3.125
Variance					6.25

Standard deviation of security B = $\sqrt{6.25} = 2.5$